Nonparametric Statistics

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Contents

1 Chapter 1: Basic Probability and Asymptotic Tools

1.1 Probability, distributions, and conditional quantities

Definition (Probability space and random variable)**.** A probability space is $(\Omega, \mathcal{A}, \mathbb{P})$. A random variable is a measurable map

$$X : \Omega \to \mathbb{R}.$$

Definition (Distribution functions)**.** The cumulative distribution function and empirical distribution function are

$$F(x) = \mathbb{P}(X \le x), \qquad F_n(x) = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}_{\{X_i \le x\}}.$$

If F is differentiable, its density is $f = F'$, and

$$F(x) = \int_{-\infty}^{x} f(t)\,dt.$$

Definition (Expectation and variance)**.** For a measurable function g,

$$\mathbb{E}[g(X)] = \int g(x)\,dF(x) = \begin{cases} \int g(x)f(x)\,dx, & X \text{ is continuous,} \\ \sum_x g(x)\mathbb{P}(X = x), & X \text{ is discrete.} \end{cases}$$

The variance of X is

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Definition (Marginal and conditional distributions)**.** For a continuous random vector $\boldsymbol{X} = (X_1, \ldots, X_p)^\top$ with joint density f,

$$f_{X_i}(x_i) = \int_{\mathbb{R}^{p-1}} f(\boldsymbol{x})\,d\boldsymbol{x}_{-i},$$

and

$$f_{X_1|\boldsymbol{X}_{-1}=\boldsymbol{x}_{-1}}(x_1) = \frac{f(\boldsymbol{x})}{f_{\boldsymbol{X}_{-1}}(\boldsymbol{x}_{-1})}.$$

The conditional mean and variance are

$$\mathbb{E}[Y \mid X] = \int y\,dF_{Y|X}(y \mid X),$$

$$\mathrm{Var}(Y \mid X) = \mathbb{E}[Y^2 \mid X] - \mathbb{E}[Y \mid X]^2.$$

Theorem (Laws of total expectation and variance)**.** *If the required moments are finite, then*

$$\mathbb{E}[Y] = \mathbb{E}\{\mathbb{E}[Y \mid X]\},$$

and

$$\mathrm{Var}(Y) = \mathbb{E}\{\mathrm{Var}(Y \mid X)\} + \mathrm{Var}\{\mathbb{E}[Y \mid X]\}.$$

Definition (Covariance and correlation)**.** For two random variables,

$$\mathrm{Cov}(X_1, X_2) = \mathbb{E}[X_1X_2] - \mathbb{E}[X_1]\mathbb{E}[X_2],$$

and

$$\mathrm{Cor}(X_1, X_2) = \frac{\mathrm{Cov}(X_1, X_2)}{\sqrt{\mathrm{Var}(X_1)\,\mathrm{Var}(X_2)}}.$$

For a random vector $\boldsymbol{X}$,

$$\mathrm{Var}(\boldsymbol{X}) = \mathbb{E}\Big[(\boldsymbol{X} - \mathbb{E}\boldsymbol{X})(\boldsymbol{X} - \mathbb{E}\boldsymbol{X})^\top\Big].$$

Theorem (Linear transformations)**.** *If A is a fixed matrix and $\boldsymbol{b}$ is fixed, then*

$$\mathbb{E}[A\boldsymbol{X} + \boldsymbol{b}] = A\mathbb{E}[\boldsymbol{X}] + \boldsymbol{b},$$

and

$$\mathrm{Var}(A\boldsymbol{X} + \boldsymbol{b}) = A\,\mathrm{Var}(\boldsymbol{X})A^\top.$$

1.2 Useful inequalities

Theorem (Markov's inequality)**.** *If $X \geq 0$, then for every $t > 0$,*

$$\mathbb{P}(X \geq t) \leq \frac{\mathbb{E}[X]}{t}.$$

Theorem (Chebyshev's inequality)**.** *If $\mathbb{E}[X] = \mu$ and $\mathrm{Var}(X) = \sigma^2 < \infty$, then*

$$\mathbb{P}(|X - \mu| \geq t) \leq \frac{\sigma^2}{t^2}, \qquad t > 0.$$

Equivalently,

$$\mathbb{P}(\mu - t\sigma < X < \mu + t\sigma) \geq 1 - \frac{1}{t^2}.$$

Theorem (Cauchy–Schwarz inequality)**.** *If $\mathbb{E}[X^2] < \infty$ and $\mathbb{E}[Y^2] < \infty$, then*

$$|\mathbb{E}[XY]| \leq \sqrt{\mathbb{E}[X^2]\mathbb{E}[Y^2]}.$$

Theorem (Jensen's inequality)**.** *If g is convex, then*

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)].$$

In particular, for $r \geq 1$,

$$|\mathbb{E}[X]|^r \leq \mathbb{E}[|X|^r].$$

If $0 < r \leq s$, then

$$\mathbb{E}|X|^s < \infty \quad \Longrightarrow \quad \mathbb{E}|X|^r < \infty.$$

1.3 Some distribution facts

Theorem (Normal distribution)**.** *For $X \sim N(\mu, \sigma^2)$,*

$$\phi_\sigma(x - \mu) = \frac{1}{\sigma}\phi\left(\frac{x - \mu}{\sigma}\right),$$

and

$$\mathbb{P}\big(X \in \mu \pm z_{\alpha/2}\sigma\big) = 1 - \alpha,$$

where

$$z_{\alpha/2} = \Phi^{-1}(1 - \alpha/2).$$

The first four moments are

$$\mathbb{E}[X] = \mu, \qquad \mathbb{E}[X^2] = \mu^2 + \sigma^2,$$
$$\mathbb{E}[X^3] = \mu^3 + 3\mu\sigma^2, \qquad \mathbb{E}[X^4] = \mu^4 + 6\mu^2\sigma^2 + 3\sigma^4.$$

Theorem (Multivariate normal distribution)**.** *For $\boldsymbol{X} \sim N_p(\boldsymbol{\mu}, \Sigma)$,*

$$\phi_\Sigma(\boldsymbol{x} - \boldsymbol{\mu}) = (2\pi)^{-p/2} |\Sigma|^{-1/2} \exp\left\{ -\frac{1}{2} (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \right\}.$$

If A is a fixed $q \times p$ matrix and $\boldsymbol{b} \in \mathbb{R}^q$, then

$$A\boldsymbol{X} + \boldsymbol{b} \sim N_q(A\boldsymbol{\mu} + \boldsymbol{b}, A\Sigma A^\top).$$

Theorem (Other distribution facts)**.** *The following facts will be used later.*

Distribution	*Main facts*
$X \sim \mathrm{LN}(\mu, \sigma^2)$	$X \overset{d}{=} \exp(Z)$ *for* $Z \sim N(\mu, \sigma^2)$, $$\mathbb{E}[X] = e^{\mu + \sigma^2/2}, \qquad \mathrm{Var}(X) = (e^{\sigma^2} - 1)e^{2\mu + \sigma^2}.$$
$X \sim \mathrm{Exp}(\lambda)$	$$f(x) = \lambda e^{-\lambda x}, \qquad x > 0.$$
$X \sim \Gamma(a, p)$	*With rate a and shape p,* $$\mathbb{E}[X] = \frac{p}{a}, \qquad \mathrm{Var}(X) = \frac{p}{a^2}.$$
$X \sim \mathrm{B}(n, p)$	$$\mathbb{E}[X] = np, \qquad \mathrm{Var}(X) = np(1-p),$$ *and* $\mathrm{B}(1, p) = \mathrm{Ber}(p)$.
$X \sim \mathrm{Pois}(\lambda)$	$$\mathbb{E}[X] = \mathrm{Var}(X) = \lambda.$$

1.4 Modes of stochastic convergence

Definition (Convergence in distribution)**.** The sequence X_n converges in distribution to X, written $X_n \xrightarrow{d} X$, if

$$F_n(x) \to F(x)$$

at every continuity point of F.

Definition (Convergence in probability)**.** The sequence X_n converges in probability to X, written $X_n \xrightarrow{\mathbb{P}} X$, if

$$\mathbb{P}(|X_n - X| > \varepsilon) \to 0$$

for every $\varepsilon > 0$.

Definition (Almost sure convergence)**.** The sequence X_n converges almost surely to X, written $X_n \xrightarrow{\text{a.s.}} X$, if

$$\mathbb{P}\{\omega : X_n(\omega) \to X(\omega)\} = 1.$$

Definition (Convergence in r-mean)**.** For $r \geq 1$, the sequence X_n converges in r-mean to X if

$$\mathbb{E}|X_n - X|^r \to 0.$$

This is written as

$$X_n \xrightarrow{r} X.$$

Theorem (Relations among modes of convergence)**.** *The basic implications are*

$$X_n \xrightarrow{r} X \Longrightarrow X_n \xrightarrow{\mathbb{P}} X,$$

$$X_n \xrightarrow{\text{a.s.}} X \Longrightarrow X_n \xrightarrow{\mathbb{P}} X \Longrightarrow X_n \xrightarrow{d} X.$$

If the limit is a constant c, then

$$X_n \xrightarrow{d} c \Longrightarrow X_n \xrightarrow{\mathbb{P}} c.$$

Definition (Mean-square consistency)**.** An estimator $\widehat{\theta}_n$ is mean-square consistent for θ if

$$\mathrm{MSE}(\widehat{\theta}_n) = \mathbb{E}[(\widehat{\theta}_n - \theta)^2] = \mathrm{Bias}(\widehat{\theta}_n)^2 + \mathrm{Var}(\widehat{\theta}_n) \to 0.$$

1.5 Main limit theorems

Theorem (Law of large numbers)**.** *If $X_1, X_2, \ldots$ are iid and $\mathbb{E}[X_i] = \mu$, then*

$$\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{\mathbb{P}} \mu,$$

and

$$\overline{X}_n \xrightarrow{\text{a.s.}} \mu.$$

Theorem (Central limit theorem)**.** *If $X_1, X_2, \ldots$ are iid with mean μ and variance $\sigma^2 < \infty$, then*

$$\frac{\sqrt{n}(\overline{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1).$$

Theorem (Lyapunov central limit theorem)**.** *Let $X_1, \ldots, X_n$ be independent with means μ_i and variances σ_i^2. Define*

$$s_n^2 = \sum_{i=1}^{n} \sigma_i^2.$$

If, for some $\delta > 0$,

$$\frac{1}{s_n^{2+\delta}} \sum_{i=1}^{n} \mathbb{E}|X_i - \mu_i|^{2+\delta} \to 0,$$

then

$$\frac{1}{s_n}\sum_{i=1}^{n}(X_i - \mu_i) \xrightarrow{d} N(0, 1).$$

Theorem (Cramér–Wold device)**.** *For random vectors $\boldsymbol{X}_n, \boldsymbol{X} \in \mathbb{R}^p$,*

$$\boldsymbol{X}_n \xrightarrow{d} \boldsymbol{X}$$

if and only if

$$\boldsymbol{c}^\top \boldsymbol{X}_n \xrightarrow{d} \boldsymbol{c}^\top \boldsymbol{X}$$

for every $\boldsymbol{c} \in \mathbb{R}^p$.

Theorem (Continuous mapping theorem)**.** *If g is continuous, then*

$$X_n \xrightarrow{d} X \quad \Longrightarrow \quad g(X_n) \xrightarrow{d} g(X),$$

$$X_n \xrightarrow{\mathbb{P}} X \quad \Longrightarrow \quad g(X_n) \xrightarrow{\mathbb{P}} g(X),$$

and

$$X_n \xrightarrow{\text{a.s.}} X \quad \Longrightarrow \quad g(X_n) \xrightarrow{\text{a.s.}} g(X).$$

Theorem (Slutsky's theorem)**.** *If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{\mathbb{P}} c$, then*

$$X_n + Y_n \xrightarrow{d} X + c,$$

$$X_n Y_n \xrightarrow{d} cX,$$

and, if $c \neq 0$,

$$\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}.$$

Theorem (Delta method)**.** *Suppose that*

$$\sqrt{n}(X_n - \mu) \xrightarrow{d} N(0, \sigma^2).$$

If g is differentiable at μ and $g'(\mu) \neq 0$, then

$$\sqrt{n}\{g(X_n) - g(\mu)\} \xrightarrow{d} N\big(0, g'(\mu)^2\sigma^2\big)\,.$$

1.6 O, o, $O_{\mathbb{P}}$, and $o_{\mathbb{P}}$

Definition (Deterministic order notation)**.** For positive deterministic sequences a_n and b_n,

$$a_n = O(b_n) \quad \Longleftrightarrow \quad \limsup_{n\to\infty} \frac{a_n}{b_n} < \infty,$$

and

$$a_n = o(b_n) \quad \Longleftrightarrow \quad \frac{a_n}{b_n} \to 0.$$

Equivalently,

$$a_n = O(b_n) \quad \Longleftrightarrow \quad \frac{a_n}{b_n} = O(1),$$

and

$$a_n = o(b_n) \quad \Longleftrightarrow \quad \frac{a_n}{b_n} = o(1).$$

Definition (Stochastic order notation)**.** For a random sequence X_n and a positive deterministic sequence a_n,

$$X_n = o_{\mathbb{P}}(a_n) \quad \Longleftrightarrow \quad \frac{|X_n|}{a_n} \xrightarrow{\mathbb{P}} 0.$$

Also,

$$X_n = O_{\mathbb{P}}(a_n) \quad \Longleftrightarrow \quad \frac{X_n}{a_n} = O_{\mathbb{P}}(1),$$

where $O_{\mathbb{P}}(1)$ means bounded in probability.

Theorem (Rules for stochastic order notation)**.** *The following relations hold:*

$$\begin{aligned}
o_{\mathbb{P}}(a_n) &\subset O_{\mathbb{P}}(a_n), \\
O_{\mathbb{P}}(a_n) + O_{\mathbb{P}}(b_n) &= O_{\mathbb{P}}(a_n + b_n), \\
o_{\mathbb{P}}(a_n) + o_{\mathbb{P}}(b_n) &= o_{\mathbb{P}}(a_n + b_n), \\
O_{\mathbb{P}}(a_n) O_{\mathbb{P}}(b_n) &= O_{\mathbb{P}}(a_n b_n), \\
o_{\mathbb{P}}(a_n) O_{\mathbb{P}}(b_n) &= o_{\mathbb{P}}(a_n b_n), \\
o_{\mathbb{P}}(1) O_{\mathbb{P}}(a_n) &= o_{\mathbb{P}}(a_n), \\
(1 + o_{\mathbb{P}}(1))^{-1} &= O_{\mathbb{P}}(1).
\end{aligned}$$

Theorem (C_p inequality)**.** *For $a, b \in \mathbb{R}$ and $p > 0$,*

$$|a + b|^p \leq C_p(|a|^p + |b|^p),$$

where

$$C_p = \begin{cases} 1, & 0 < p \leq 1, \\ 2^{p-1}, & p > 1. \end{cases}$$

Theorem (Order from the variance)**.** *If $\mathrm{Var}(X_n) < \infty$, then*

$$X_n - \mathbb{E}[X_n] = O_{\mathbb{P}}\Big(\sqrt{\mathrm{Var}(X_n)}\Big).$$

Consequently,

$$X_n = \mathbb{E}[X_n] + O_{\mathbb{P}}\Big(\sqrt{\mathrm{Var}(X_n)}\Big).$$

1.7 Analytical tools

Theorem (Mean value theorem)**.** *If f is continuous on $[a, b]$ and differentiable on (a, b), then*

$$f(b) - f(a) = f'(c)(b - a)$$

for some $c \in (a, b)$.

Theorem (Integral mean value theorem)**.** *If f is continuous on $[a, b]$, then*

$$\int_a^b f(x)\, dx = f(c)(b - a)$$

for some $c \in (a, b)$.

Theorem (Taylor's theorem)**.** *If f has p continuous derivatives near x, then*

$$f(x + h) = \sum_{j=0}^{p} \frac{f^{(j)}(x)}{j!} h^j + o(h^p).$$

If $f^{(p+1)}$ is bounded locally, the remainder can be controlled uniformly in a neighborhood of x.

Theorem (Dominated convergence theorem)**.** *Suppose that*

$$f_n(x) \to f(x), \qquad |f_n(x)| \leq g(x),$$

and

$$\int |g(x)|\, dx < \infty.$$

Then

$$\lim_{n\to\infty} \int f_n(x)\, dx = \int f(x)\, dx.$$

Remark (Parametric and nonparametric inference). Parametric methods obtain efficiency by imposing a finite-dimensional model. Nonparametric methods use weaker assumptions and remain valid over a wider class of distributions. The usual trade-off is

$$\text{stronger model assumptions}$$
$$\Updownarrow$$
$$\text{more efficiency, but less protection against misspecification.}$$

2 Chapter 2: Univariate Kernel Density Estimation

Let $X_1, \dots, X_n$ be iid from a continuous density f with cdf F.

2.1 Histogram

Since

$$f(x_0) = F'(x_0) = \lim_{h \downarrow 0} \frac{F(x_0 + h) - F(x_0)}{h} = \lim_{h \downarrow 0} \frac{\mathbb{P}(x_0 < X < x_0 + h)}{h},$$

a histogram estimates local probability per unit length.

Choose an origin t_0 and a bin width $h > 0$. Define

$$t_k = t_0 + hk, \qquad B_k = [t_k, t_{k+1}), \qquad k \in \mathbb{Z}.$$

For $x \in B_k$,

$$\widehat{f}_H(x; t_0, h) = \frac{1}{nh} \sum_{i=1}^{n} \mathbf{1}_{\{X_i \in B_k\}} = \frac{v_k}{nh},$$

where $v_k = \#\{i : X_i \in B_k\}$.

Let

$$p_k = \mathbb{P}(X \in B_k) = \int_{B_k} f(t)\, dt.$$

Then

$$v_k \sim \mathrm{B}(n, p_k).$$

If f is continuous, the integral mean value theorem gives

$$p_k = hf(\xi_{k,h}) \quad \text{for some } \xi_{k,h} \in (t_k, t_{k+1}).$$

Therefore,

$$\mathbb{E}[\widehat{f}_H(x; t_0, h)] = f(\xi_{k,h}),$$

$$\mathrm{Var}\{\widehat{f}_H(x; t_0, h)\} = \frac{f(\xi_{k,h})\{1 - hf(\xi_{k,h})\}}{nh}.$$

If $h \to 0$, then $\xi_{k,h} \to x$ and the bias disappears. To make the variance vanish, one also needs

$$nh \to \infty.$$

The histogram depends on both t_0 and h.

2.2 Moving histogram

Using the symmetric derivative,

$$f(x) = \lim_{h \downarrow 0} \frac{F(x+h) - F(x-h)}{2h} = \lim_{h \downarrow 0} \frac{\mathbb{P}(x-h < X < x+h)}{2h}.$$

The moving histogram, also called the naive density estimator, is

$$\widehat{f}_N(x;h) = \frac{1}{2nh} \sum_{i=1}^{n} \mathbf{1}_{\{x-h<X_i<x+h\}}.$$

Let

$$p_{x,h} = F(x+h) - F(x-h).$$

Then

$$\sum_{i=1}^{n} \mathbf{1}_{\{x-h<X_i<x+h\}} \sim \mathrm{B}(n, p_{x,h}),$$

so

$$\mathbb{E}[\widehat{f}_N(x;h)] = \frac{F(x+h) - F(x-h)}{2h},$$

$$\mathrm{Var}\{\widehat{f}_N(x;h)\} = \frac{F(x+h) - F(x-h)}{4nh^2} - \frac{\{F(x+h) - F(x-h)\}^2}{4nh^2}.$$

For small h,

$$\mathrm{Var}\{\widehat{f}_N(x;h)\} \approx \frac{f(x)}{2nh} - \frac{f(x)^2}{n}.$$

Thus the basic consistency conditions are again

$$h \to 0, \qquad nh \to \infty.$$

The moving histogram is the kernel estimator obtained from the rectangular kernel

$$K(z) = \frac{1}{2} \mathbf{1}_{\{|z|<1\}}.$$

It uses local data, but all observations inside $(x-h, x+h)$ receive the same weight.

2.3 Kernel density estimator

Define the scaled kernel

$$K_h(z) = \frac{1}{h} K\left(\frac{z}{h}\right).$$

The kernel density estimator is

$$\widehat{f}(x;h) = \frac{1}{n} \sum_{i=1}^{n} K_h(x - X_i) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - X_i}{h}\right).$$

Usually K is a symmetric density with its mode at zero.

Common kernels are

$$\begin{aligned} \text{Normal:} \qquad & K(z) = \phi(z), \\ \text{Epanechnikov:} \qquad & K(z) = \frac{3}{4}(1 - z^2)\mathbf{1}_{\{|z|<1\}}, \end{aligned}$$

$$\text{Rectangular:} \qquad K(z) = \frac{1}{2}\mathbf{1}_{\{|z|<1\}}.$$

For the normal kernel,

$$K_h(x - X_i) = \phi_h(x - X_i),$$

so $\widehat{f}$ is an equally weighted mixture of $N(X_i, h^2)$ densities.

Useful notation:

$$\mu_2(K) = \int z^2 K(z)\,dz, \qquad R(g) = \int g(x)^2\,dx,$$

$$(f * g)(x) = \int f(x - y)g(y)\,dy.$$

Once the kernel is symmetric and regular enough, the choice of h matters much more than the exact shape of K.

2.4 Assumptions for the asymptotic analysis

A1. f is square integrable, twice continuously differentiable, and f'' is square integrable.

A2. K is a symmetric and bounded density, has finite second moment, and is square integrable.

A3. $h = h_n$ is deterministic, $h \to 0$, and $nh \to \infty$.

2.5 Exact expectation and variance

By linearity and convolution,

$$\mathbb{E}[\widehat{f}(x;h)] = \int K_h(x - y)f(y)\,dy = (K_h * f)(x),$$

$$\mathrm{Var}\{\widehat{f}(x;h)\} = \frac{1}{n}\left\{(K_h^2 * f)(x) - (K_h * f)(x)^2\right\}.$$

The estimator is biased for fixed h because $(K_h * f)(x) \neq f(x)$ in general.

2.6 Asymptotic bias, variance, and MSE

Using $z = (x - y)/h$,

$$\mathbb{E}[\widehat{f}(x;h)] = \int K(z)f(x - hz)\,dz.$$

A second-order Taylor expansion gives

$$f(x - hz) = f(x) - hf'(x)z + \frac{1}{2}h^2 f''(x)z^2 + o(h^2z^2).$$

Because K is symmetric,

$$\int zK(z)\,dz = 0.$$

Hence, under A1–A3,

$$\mathrm{Bias}\{\widehat{f}(x;h)\} = \frac{1}{2}\mu_2(K)f''(x)h^2 + o(h^2),$$

$$\mathrm{Var}\{\widehat{f}(x;h)\} = \frac{R(K)}{nh}f(x) + o((nh)^{-1}).$$

Therefore,

$$\mathrm{MSE}\{\widehat{f}(x;h)\} = \frac{\mu_2(K)^2}{4} f''(x)^2 h^4 + \frac{R(K)}{nh} f(x) + o\big(h^4 + (nh)^{-1}\big)\,.$$

Equivalently,

$$\widehat{f}(x;h) = f(x) + O(h^2) + O_{\mathbb{P}}((nh)^{-1/2}).$$

Thus $\widehat{f}(x;h) \xrightarrow{\mathbb{P}} f(x)$ under A1–A3.

Here are some interpretations of the bias.

- If $f''(x) < 0$, then f is locally concave and the leading bias is negative. Peaks tend to be underestimated.
- If $f''(x) > 0$, then f is locally convex and the leading bias is positive. Valleys and tails tend to be overestimated.
- Larger curvature means a harder density estimation problem.

Also, we can interpret the variance as below.

$$\mathrm{Var}\{\widehat{f}(x;h)\} \approx \frac{R(K)}{nh} f(x).$$

The quantity nh acts as an effective sample size.

2.7 Asymptotic normality

Assume additionally that

$$\int K(z)^{2+\delta}\,dz < \infty \quad \text{for some } \delta > 0.$$

Then

$$\sqrt{nh}\left\{\widehat{f}(x;h) - \mathbb{E}[\widehat{f}(x;h)]\right\} \xrightarrow{d} N\big(0, R(K)f(x)\big).$$

If also $nh^5 = O(1)$, then

$$\sqrt{nh}\left\{\widehat{f}(x;h) - f(x) - \frac{1}{2}\mu_2(K)f''(x)h^2\right\} \xrightarrow{d} N\big(0, R(K)f(x)\big).$$

The rate is $\sqrt{nh}$, not $\sqrt{n}$.

2.8 Integrated error criteria

$$\mathrm{ISE}\{\widehat{f}(\cdot;h)\} = \int \{\widehat{f}(x;h) - f(x)\}^2\,dx,$$

$$\mathrm{MISE}\{\widehat{f}(\cdot;h)\} = \mathbb{E}[\mathrm{ISE}\{\widehat{f}(\cdot;h)\}] = \int \mathrm{MSE}\{\widehat{f}(x;h)\}\,dx,$$

$$\mathrm{MIAE}\{\widehat{f}(\cdot;h)\} = \mathbb{E}\int |\widehat{f}(x;h) - f(x)|\,dx.$$

The integrated absolute error is invariant under monotone transformations, but the squared error is easier to analyze.

Integrating the pointwise MSE gives

$$\mathrm{MISE}\{\widehat{f}(\cdot;h)\} = \frac{1}{4}\mu_2(K)^2 R(f'')h^4 + \frac{R(K)}{nh} + o\big(h^4 + (nh)^{-1}\big)\,.$$

The leading part is

$$\mathrm{AMISE}\{\widehat{f}(\cdot;h)\} = \frac{1}{4}\mu_2(K)^2 R(f'')h^4 + \frac{R(K)}{nh}.$$

The minimizing bandwidth is

$$h_{\mathrm{AMISE}} = \left\{\frac{R(K)}{\mu_2(K)^2 R(f'')n}\right\}^{1/5}.$$

The optimal order is

$$\inf_{h>0} \mathrm{AMISE}\{\widehat{f}(\cdot;h)\} = O(n^{-4/5}), \qquad h_{\mathrm{AMISE}} = O(n^{-1/5}).$$

2.9 Bandwidth selection

2.9.1 Normal scale and rule of thumb

Under the normal kernel density,

$$R\{\phi''_\sigma(\cdot - \mu)\} = \frac{3}{8\sqrt{\pi}\,\sigma^5}.$$

Thus

$$\widehat{h}_{\mathrm{NS}} = \left\{\frac{8\sqrt{\pi}R(K)}{3\mu_2(K)^2 n}\right\}^{1/5}\widehat{\sigma}.$$

A robust scale estimate is

$$\widehat{\sigma} = \min(s, \widehat{\sigma}_{\mathrm{IQR}}), \qquad \widehat{\sigma}_{\mathrm{IQR}} = \frac{X_{[0.75n]} - X_{[0.25n]}}{\Phi^{-1}(0.75) - \Phi^{-1}(0.25)}.$$

For the normal kernel, $\mu_2(K) = 1$ and $R(K) = 1/(2\sqrt{\pi})$, so

$$\widehat{h}_{\mathrm{RT}} = \left(\frac{4}{3}\right)^{1/5} n^{-1/5}\widehat{\sigma} \approx 1.06n^{-1/5}\widehat{\sigma}.$$

This selector is simple but may oversmooth a density that is far from normal.

2.9.2 Direct plug-in selector

Repeated integration by parts gives

$$R(f^{(s)}) = (-1)^s \int f^{(2s)}(x)f(x)\,dx.$$

Define

$$\psi_r = \int f^{(r)}(x)f(x)\,dx = \mathbb{E}[f^{(r)}(X)].$$

Then $R(f'') = \psi_4$. A kernel estimator of ψ_r is

$$\widehat{\psi}_r(g) = \frac{1}{n}\sum_{i=1}^n \widehat{f}^{(r)}(X_i;g) = \frac{1}{n^2}\sum_{i=1}^n\sum_{j=1}^n L_g^{(r)}(X_i - X_j).$$

For a normal kernel and even r,

$$\psi_r = \frac{(-1)^{r/2} r!}{(2\sigma)^{r+1}(r/2)!\sqrt{\pi}}.$$

The usual two-stage DPI construction is

$$\widehat{\psi}_8^{\mathrm{NS}} = \frac{105}{32\sqrt{\pi}\,\widehat{\sigma}^9},$$

$$g_1 = \left\{ -\frac{2K^{(6)}(0)}{\mu_2(K)\widehat{\psi}_8^{\mathrm{NS}} n} \right\}^{1/9}, \qquad \widehat{\psi}_6(g_1),$$

$$g_2 = \left\{ -\frac{2K^{(4)}(0)}{\mu_2(K)\widehat{\psi}_6(g_1) n} \right\}^{1/7}, \qquad \widehat{\psi}_4(g_2),$$

$$\widehat{h}_{\mathrm{DPI}} = \left\{ \frac{R(K)}{\mu_2(K)^2 \widehat{\psi}_4(g_2) n} \right\}^{1/5}.$$

2.9.3 Least-squares cross-validation

Expand the MISE as

$$\mathrm{MISE} = \mathbb{E}\int \widehat{f}^2 - 2\mathbb{E}\int \widehat{f} f + \int f^2.$$

The last term does not depend on h. Define the leave-one-out estimator

$$\widehat{f}_{-i}(x;h) = \frac{1}{n-1}\sum_{j\neq i} K_h(x - X_j).$$

The least-squares cross-validation criterion is

$$\mathrm{LSCV}(h) = \int \widehat{f}(x;h)^2\,dx - \frac{2}{n}\sum_{i=1}^{n} \widehat{f}_{-i}(X_i;h).$$

It satisfies

$$\mathbb{E}[\mathrm{LSCV}(h)] = \mathrm{MISE}\{\widehat{f}(\cdot;h)\} - R(f).$$

Thus

$$\widehat{h}_{\mathrm{LSCV}} = \arg\min_{h>0} \mathrm{LSCV}(h).$$

The objective may have several local minima, so a grid check is useful.

2.9.4 Biased cross-validation

A leave-out-diagonal estimate of $R(f'')$ is

$$\widehat{R(f'')} = R\{\widehat{f}''(\cdot;h)\} - \frac{R(K'')}{nh^5} = \frac{1}{n^2}\sum_{i=1}^{n}\sum_{j\neq i}(K_h'' * K_h'')(X_i - X_j).$$

Then

$$\mathrm{BCV}(h) = \frac{1}{4}\mu_2(K)^2 \widehat{R(f'')} h^4 + \frac{R(K)}{nh},$$

$$\widehat{h}_{\mathrm{BCV}} = \arg\operatorname*{locmin}_{h>0} \mathrm{BCV}(h),$$

where the smallest local minimizer is used. BCV is usually less variable but more biased than LSCV.

2.9.5 Comparison

Under regularity conditions,

$$n^{1/10}\left(\frac{\widehat{h}_{\text{LSCV}}}{h_{\text{MISE}}}-1\right)\xrightarrow{d}N(0,\sigma^2_{\text{LSCV}}),$$

$$n^{1/10}\left(\frac{\widehat{h}_{\text{BCV}}}{h_{\text{MISE}}}-1\right)\xrightarrow{d}N(0,\sigma^2_{\text{BCV}}),$$

$$n^{5/14}\left(\frac{\widehat{h}_{\text{DPI}}}{h_{\text{MISE}}}-1\right)\xrightarrow{d}N(0,\sigma^2_{\text{DPI}}).$$

Thus DPI converges faster. LSCV is flexible but variable. BCV is more stable and tends to choose a larger bandwidth.

2.10 Boundary correction by transformation

If the support has a boundary, an ordinary kernel spreads mass outside the support. Near a boundary, the bias is generally of order $O(h)$ rather than $O(h^2)$.

Let $Y = t(X)$ have density g. Then

$$f(x) = g(t(x))t'(x).$$

Estimate g on the transformed scale and return to the original scale:

$$\widehat{f}_T(x;h,t) = \frac{1}{n}\sum_{i=1}^{n} K_h\{t(x) - t(X_i)\}t'(x).$$

Typical choices are the log transformation for (a,∞) and the probit transformation for (a,b).

2.11 Sampling from a KDE

Since

$$\widehat{f}(x;h) = \frac{1}{n}\sum_{i=1}^{n} K_h(x - X_i),$$

sampling from $\widehat{f}$ is sampling from a finite mixture:

1. choose I uniformly from $\{1,\ldots,n\}$;
2. sample $Z \sim K$ independently;
3. set $X^* = X_I + hZ$.

3 Chapter 4: Univariate Kernel Regression

Let $(X_1,Y_1),\ldots,(X_n,Y_n)$ be iid observations. The target is the regression function

$$m(x) = \mathbb{E}[Y \mid X = x].$$

If (X,Y) is continuous, then

$$m(x) = \int y f_{Y|X=x}(y)\,dy = \frac{\int y f(x,y)\,dy}{f_X(x)}.$$

3.1 Nadaraya–Watson estimator

Use a product kernel estimate of the joint density,

$$\widehat{f}(x, y; h_1, h_2) = \frac{1}{n}\sum_{i=1}^{n} K_{h_1}(x - X_i)K_{h_2}(y - Y_i),$$

and

$$\widehat{f}_X(x; h_1) = \frac{1}{n}\sum_{i=1}^{n} K_{h_1}(x - X_i).$$

Then

$$\begin{aligned}\frac{\int y\widehat{f}(x, y; h_1, h_2)\,dy}{\widehat{f}_X(x; h_1)} &= \frac{\sum_{i=1}^{n} K_{h_1}(x - X_i)\int yK_{h_2}(y - Y_i)\,dy}{\sum_{j=1}^{n} K_{h_1}(x - X_j)}\\ &= \frac{\sum_{i=1}^{n} K_{h_1}(x - X_i)Y_i}{\sum_{j=1}^{n} K_{h_1}(x - X_j)}.\end{aligned}$$

The bandwidth in the Y direction disappears. Writing $h = h_1$,

$$\widehat{m}(x; 0, h) = \sum_{i=1}^{n} W_i^0(x)Y_i, \qquad W_i^0(x) = \frac{K_h(x - X_i)}{\sum_{j=1}^{n} K_h(x - X_j)}.$$

Since $W_i^0(x) \geq 0$ and $\sum_i W_i^0(x) = 1$, this is a local weighted mean. It is also called the local constant estimator.

3.2 Local polynomial estimator

For X_i near x,

$$m(X_i) \approx m(x) + m'(x)(X_i - x) + \cdots + \frac{m^{(p)}(x)}{p!}(X_i - x)^p.$$

Replace the unknown derivatives by coefficients $\beta_0, \ldots, \beta_p$ and solve

$$\widehat{\boldsymbol{\beta}}_h = \underset{\boldsymbol{\beta}\in\mathbb{R}^{p+1}}{\arg\min} \sum_{i=1}^{n} \left\{ Y_i - \sum_{j=0}^{p} \beta_j (X_i - x)^j \right\}^2 K_h(x - X_i).$$

Define

$$\mathsf{X}_x = \begin{pmatrix} 1 & X_1 - x & \cdots & (X_1 - x)^p \\ \vdots & \vdots & & \vdots \\ 1 & X_n - x & \cdots & (X_n - x)^p \end{pmatrix},$$

$$\mathsf{W}_x = \mathrm{diag}\{K_h(X_1 - x), \ldots, K_h(X_n - x)\}, \qquad \boldsymbol{Y} = (Y_1, \ldots, Y_n)^\top.$$

Then

$$\widehat{\boldsymbol{\beta}}_h = (\mathsf{X}_x^\top \mathsf{W}_x \mathsf{X}_x)^{-1}\mathsf{X}_x^\top \mathsf{W}_x \boldsymbol{Y}.$$

The local polynomial estimate is the fitted intercept:

$$\widehat{m}(x; p, h) = \widehat{\beta}_{h,0} = \boldsymbol{e}_1^\top (\mathsf{X}_x^\top \mathsf{W}_x \mathsf{X}_x)^{-1}\mathsf{X}_x^\top \mathsf{W}_x \boldsymbol{Y} = \sum_{i=1}^{n} W_i^p(x)Y_i.$$

For $p = 0$, this reduces to Nadaraya–Watson. For $p = 1$, define

$$\widehat{s}_r(x; h) = \frac{1}{n} \sum_{i=1}^{n} (X_i - x)^r K_h(x - X_i), \qquad r = 0, 1, 2.$$

Then the local linear weights are

$$W_i^1(x) = \frac{1}{n} \frac{\widehat{s}_2(x; h) - \widehat{s}_1(x; h)(X_i - x)}{\widehat{s}_2(x; h)\widehat{s}_0(x; h) - \widehat{s}_1(x; h)^2} K_h(x - X_i).$$

The weights sum to one, but they need not be nonnegative.

3.3 Regression model and assumptions

Write the location–scale model as

$$Y = m(X) + \sigma(X)\varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0, \qquad \operatorname{Var}(\varepsilon \mid X) = 1,$$

so that

$$\sigma^2(x) = \operatorname{Var}(Y \mid X = x).$$

Let f denote the marginal density of X.

A1. m is twice continuously differentiable.

A2. σ^2 is continuous and positive.

A3. f is continuously differentiable and bounded away from zero on the region of interest.

A4. K is a symmetric and bounded density, has finite second moment, and is square integrable.

A5. $h = h_n$ is deterministic, $h \to 0$, and $nh \to \infty$.

3.4 Conditional bias and variance

Condition on $X_1, \ldots, X_n$. For $p = 0, 1$,

$$\operatorname{Bias}\{\widehat{m}(x; p, h) \mid X_1, \ldots, X_n\} = B_p(x)h^2 + o_{\mathbb{P}}(h^2),$$

$$\operatorname{Var}\{\widehat{m}(x; p, h) \mid X_1, \ldots, X_n\} = \frac{R(K)}{nhf(x)}\sigma^2(x) + o_{\mathbb{P}}((nh)^{-1}),$$

where

$$B_0(x) = \frac{\mu_2(K)}{2} \left\{ m''(x) + 2\frac{m'(x)f'(x)}{f(x)} \right\},$$

$$B_1(x) = \frac{\mu_2(K)}{2} m''(x).$$

Local constant versus local linear.

- Both have bias of order h^2 and variance of order $(nh)^{-1}$.
- The local constant bias contains the extra design term $2m'(x)f'(x)/f(x)$.
- The local linear estimator removes this term and has the same first-order variance.
- A local constant fit is exactly unbiased for a constant regression function.
- A local linear fit is exactly unbiased for a linear regression function.

- More generally, an odd local polynomial order is usually preferred to the preceding even order.

The variance has the form

$$\mathrm{Var}\{\widehat{m}(x;p,h) \mid X_1,\dots,X_n\} \approx \frac{R(K)\sigma^2(x)}{nhf(x)}.$$

Thus low predictor density and large conditional variance both increase uncertainty.

3.5 Asymptotic normality

Assume that for some $\delta > 0$,

$$\mathbb{E}\{|Y - m(x)|^{2+\delta} \mid X = x\} < \infty.$$

Then, under A1–A5,

$$\sqrt{nh}\,\{\widehat{m}(x;p,h) - \mathbb{E}[\widehat{m}(x;p,h) \mid X_1,\dots,X_n]\} \xrightarrow{d} N\left(0, \frac{R(K)\sigma^2(x)}{f(x)}\right).$$

If also $nh^5 = O(1)$, then

$$\sqrt{nh}\,\{\widehat{m}(x;p,h) - m(x) - B_p(x)h^2\} \xrightarrow{d} N\left(0, \frac{R(K)\sigma^2(x)}{f(x)}\right).$$

3.6 Bandwidth selection for local polynomial regression

A density-weighted integrated squared error is

$$\mathrm{ISE}\{\widehat{m}(\cdot;p,h)\} = \int \{\widehat{m}(x;p,h) - m(x)\}^2 f(x)\,dx.$$

Conditioning on the predictors,

$$\mathrm{MISE}\{\widehat{m}(\cdot;p,h) \mid X_1,\dots,X_n\} = \int \mathrm{MSE}\{\widehat{m}(x;p,h) \mid X_1,\dots,X_n\} f(x)\,dx.$$

The leading asymptotic error is

$$\mathrm{AMISE}\{\widehat{m}(\cdot;p,h) \mid X_1,\dots,X_n\} = h^4 \int B_p(x)^2 f(x)\,dx + \frac{R(K)}{nh} \int \sigma^2(x)\,dx.$$

For the local linear estimator,

$$\theta_{22}(m) = \int m''(x)^2 f(x)\,dx,$$

so

$$h_{\mathrm{AMISE}} = \left\{ \frac{R(K)\int \sigma^2(x)\,dx}{\mu_2(K)^2\theta_{22}(m)n} \right\}^{1/5}.$$

3.6.1 Rule-of-thumb selector

Fit a global quartic model

$$\widehat{m}_Q(x) = \widehat{\alpha}_0 + \widehat{\alpha}_1 x + \widehat{\alpha}_2 x^2 + \widehat{\alpha}_3 x^3 + \widehat{\alpha}_4 x^4.$$

Then

$$\widehat{m}_Q''(x) = 2\widehat{\alpha}_2 + 6\widehat{\alpha}_3 x + 12\widehat{\alpha}_4 x^2,$$

$$\widehat{\theta}_{22}(\widehat{m}_Q) = \frac{1}{n}\sum_{i=1}^{n} \widehat{m}_Q''(X_i)^2.$$

Under homoscedasticity,

$$\widehat{\sigma}_Q^2 = \frac{1}{n-5}\sum_{i=1}^{n}\{Y_i - \widehat{m}_Q(X_i)\}^2,$$

and approximate

$$\int \sigma^2(x)\,dx \approx \{X_{(n)} - X_{(1)}\}\widehat{\sigma}_Q^2.$$

This gives

$$\widehat{h}_{\mathrm{RT}} = \left\{ \frac{R(K)\{X_{(n)} - X_{(1)}\}\widehat{\sigma}_Q^2}{\mu_2(K)^2 \widehat{\theta}_{22}(\widehat{m}_Q) n} \right\}^{1/5}.$$

Its quality depends on the quartic approximation and the homoscedasticity assumption.

3.6.2 Cross-validation

The ordinary residual criterion

$$\frac{1}{n}\sum_{i=1}^{n}\{Y_i - \widehat{m}(X_i; p, h)\}^2$$

favors $h \approx 0$ because a very small bandwidth can nearly interpolate the data.
Instead, define the leave-one-out estimator $\widehat{m}_{-i}$ and

$$\mathrm{CV}(h) = \frac{1}{n}\sum_{i=1}^{n}\{Y_i - \widehat{m}_{-i}(X_i; p, h)\}^2, \qquad \widehat{h}_{\mathrm{CV}} = \arg\min_{h>0} \mathrm{CV}(h).$$

Because $\widehat{m}$ is linear in the responses,

$$W_{-i,j}^p(x) = \frac{W_j^p(x)}{1 - W_i^p(x)}, \qquad j \neq i,$$

and

$$\mathrm{CV}(h) = \frac{1}{n}\sum_{i=1}^{n}\left\{ \frac{Y_i - \widehat{m}(X_i; p, h)}{1 - W_i^p(X_i)} \right\}^2.$$

The CV objective may have local minima, so several starting values or a grid search are useful.

3.7 Naive regression estimator and regressogram

With the rectangular kernel,

$$\widehat{m}_N(x;h) = \frac{\sum_{i=1}^n \mathbf{1}_{\{|X_i-x|<h\}} Y_i}{\sum_{i=1}^n \mathbf{1}_{\{|X_i-x|<h\}}} = \frac{1}{|\mathcal{N}(x;h)|} \sum_{i\in\mathcal{N}(x;h)} Y_i,$$

where

$$\mathcal{N}(x;h) = \{i : |X_i - x| < h\}.$$

It is undefined when $|\mathcal{N}(x;h)| = 0$ and is generally discontinuous.

For fixed bins $B_k = [t_k, t_{k+1})$,

$$\widehat{m}_R(x;h) = \frac{\sum_{i=1}^n \mathbf{1}_{\{X_i\in B_k\}} Y_i}{\sum_{i=1}^n \mathbf{1}_{\{X_i\in B_k\}}} = \frac{1}{|\mathcal{B}(k;h)|} \sum_{i\in\mathcal{B}(k;h)} Y_i, \qquad x \in B_k.$$

This is the regressogram. It is constant on each fixed bin and is more rigid than the moving-window estimator.

3.8 Behavior in low-density regions

Both local constant and local linear estimators can be unstable where $f(x)$ is small or where the support of X has a gap. The local constant fit tends to stay near the response attached to the closest observation. The local linear fit extrapolates a local line and may produce large spikes. Nonparametric regression should therefore be interpreted mainly on regions supported by the predictor data.

4 Appendix A: Pointwise Confidence Intervals for a Density

4.1 Why a direct interval for $f(x)$ is difficult

For fixed h,

$$\mathbb{E}[\widehat{f}(x;h)] = (K_h * f)(x),$$

not $f(x)$. The difference

$$(K_h * f)(x) - f(x)$$

is the smoothing bias. Removing its leading term requires estimating $f''(x)$, which is harder than estimating $f(x)$ itself.

Therefore, the simple interval below targets the smoothed density

$$\mathbb{E}[\widehat{f}(x;h)] = (K_h * f)(x),$$

not the original density $f(x)$.

4.2 Normal convolution example

For normal densities,

$$\{\phi_{\sigma_1}(\cdot - \mu_1) * \phi_{\sigma_2}(\cdot - \mu_2)\}(x) = \phi_{\sqrt{\sigma_1^2+\sigma_2^2}}(x - \mu_1 - \mu_2).$$

Also,

$$\int \phi_{\sigma_1}(x-\mu_1)\phi_{\sigma_2}(x-\mu_2)\,dx = \phi_{\sqrt{\sigma_1^2+\sigma_2^2}}(\mu_1-\mu_2).$$

If $K=\phi$ and $f(x)=\phi_\sigma(x-\mu)$, then

$$\mathbb{E}[\widehat{f}(x;h)] = (K_h * f)(x) = \phi_{\sqrt{h^2+\sigma^2}}(x-\mu).$$

Thus the mean of the KDE is the density of $N(\mu,\sigma^2+h^2)$.

The exact variance is

$$\mathrm{Var}\{\widehat{f}(x;h)\} = \frac{1}{n}\left\{\frac{1}{2\sqrt{\pi}h}\phi_{\sqrt{h^2/2+\sigma^2}}(x-\mu) - \phi_{\sqrt{h^2+\sigma^2}}(x-\mu)^2\right\}.$$

4.3 Asymptotic pointwise interval

From the KDE central limit theorem,

$$\sqrt{nh}\{\widehat{f}(x;h) - \mathbb{E}[\widehat{f}(x;h)]\} \xrightarrow{d} N(0, R(K)f(x)).$$

Since

$$\widehat{f}(x;h) = f(x)\{1+o_{\mathbb{P}}(1)\},$$

Slutsky's theorem gives

$$\sqrt{\frac{nh}{R(K)\widehat{f}(x;h)}}\{\widehat{f}(x;h) - \mathbb{E}[\widehat{f}(x;h)]\} \xrightarrow{d} N(0,1).$$

Hence an asymptotic pointwise $100(1-\alpha)\%$ interval for the smoothed density is

$$I_\alpha(x) = \left[\widehat{f}(x;h) \pm z_{\alpha/2}\sqrt{\frac{R(K)\widehat{f}(x;h)}{nh}}\right].$$

Important remarks.

1. The target is $(K_h * f)(x)$, not $f(x)$.
2. The interval is pointwise:

   $$\mathbb{P}\{(K_h * f)(x) \in I_\alpha(x)\} \approx 1-\alpha$$

   for a fixed x. This does not give simultaneous coverage over all x.
3. Good coverage needs both small bias and enough local data: h should be small and nh should be large.
4. The derivation treats h as deterministic. A data-driven bandwidth can change finite-sample coverage.
5. Replacing $f(x)$ by $\widehat{f}(x;h)$ is least reliable in very low-density regions.

For the normal kernel case, one may benchmark the interval above with the interval based on the exact variance,

$$\widehat{f}(x;h) \pm z_{\alpha/2}\sqrt{\mathrm{Var}\{\widehat{f}(x;h)\}}.$$